ArteryAI - Segmentation Performance Report

Comparison of Baseline U-Net and Attention U-Net

Project Objective

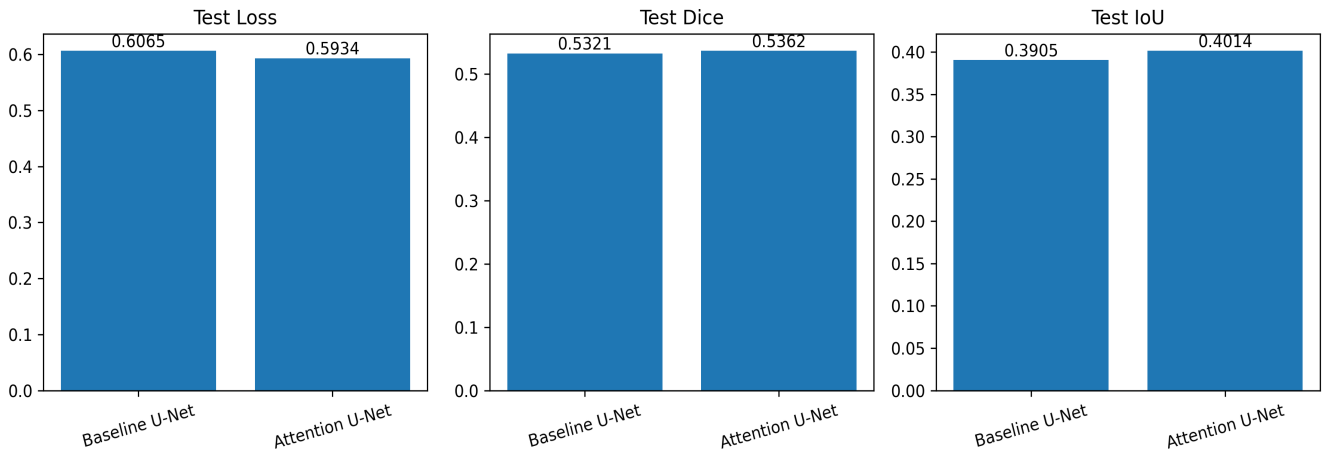
This report evaluates two deep-learning architectures for coronary anatomy segmentation using the ARCADE syntax dataset. The evaluated models are a baseline U-Net and an Attention U-Net.

Experimental Configuration

Setting	Value
Dataset	ARCADE Syntax
Training images	1000
Validation images	200
Test images	300
Image resolution	128 x 128
Batch size	1
Training epochs	5
Optimizer	Adam
Learning rate	0.001
Loss function	Dice + Binary Cross Entropy
Device	CPU

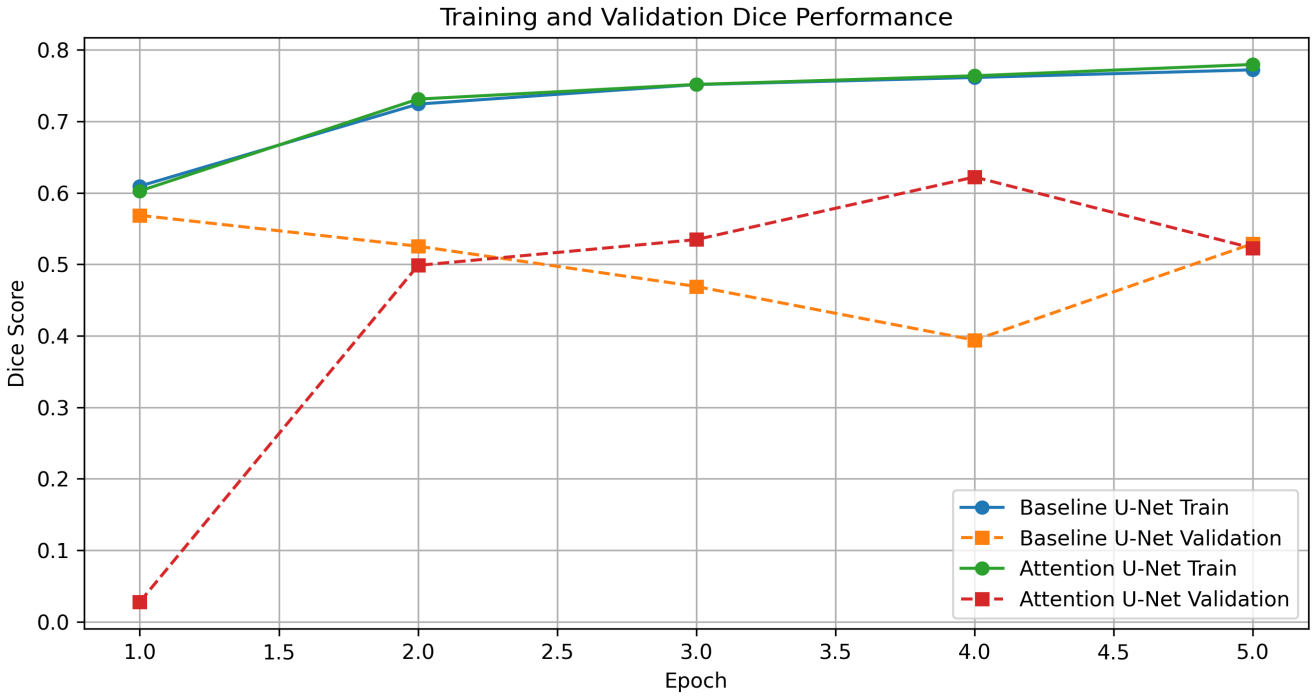
Test Performance Comparison

Model	Test Loss	Dice	IoU
Baseline U-Net	0.6065	0.5321	0.3905
Attention U-Net	0.5934	0.5362	0.4014



Training and Validation Analysis

The graph presents the Dice scores recorded during training and validation for both architectures. Differences between training and validation performance may indicate generalization challenges.



Best Validation Checkpoints

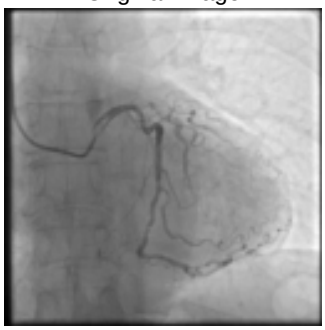
Model	Best Epoch	Validation Dice
Baseline U-Net	1	0.5685
Attention U-Net	4	0.6222

Segmentation Prediction Examples

The following are actual saved outputs from the trained models. Each set contains the original image, ground-truth mask and predicted mask.

Baseline U-Net - Example 1

Original Image



Ground Truth



Prediction



Baseline U-Net - Example 2

Original Image



Ground Truth

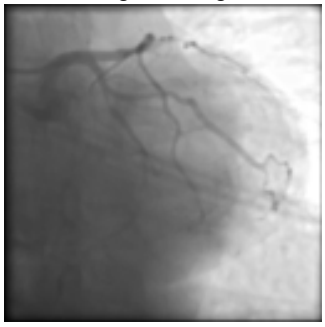


Prediction



Baseline U-Net - Example 3

Original Image



Ground Truth



Prediction



Attention U-Net - Example 1

Original Image



Ground Truth

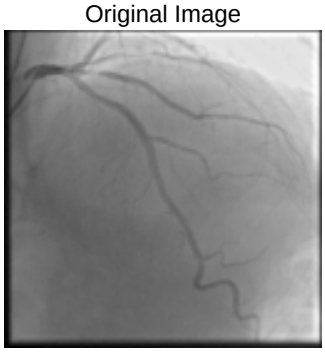


Prediction



Attention U-Net - Example 2

Original Image



Ground Truth

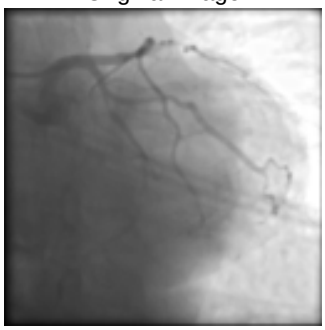


Prediction



Attention U-Net - Example 3

Original Image



Ground Truth



Prediction



Observations

Attention U-Net test Dice difference: +0.0041. Test IoU difference: +0.0109. Test loss reduction: +0.0130. Attention U-Net achieved slightly better results on all three test metrics in this experiment.

Limitations

- Both models were trained for only five epochs.

- Training was performed on CPU at 128 x 128 resolution.

- Results are based on one training run and may vary with random seeds.

- Validation Dice scores fluctuated across epochs.

- The binary mask combines syntax polygons and excludes stenosis. This target definition should be checked against the dataset protocol.

- These experimental results are not clinical validation and must not be used for diagnosis or treatment decisions.

Conclusion

Both U-Net architectures completed training and held-out test evaluation. Attention U-Net achieved slightly better test loss, Dice and IoU in this experiment. Further experiments with additional epochs, repeated runs and careful validation are recommended before drawing stronger conclusions.